

# Towards Objective and Implicit Presence Measures in Mixed Reality

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Figure 1: An overview of our approach to assessing using an implicit cue and interaction mechanisms to estimate reaction time.

## Abstract

Measuring and quantifying presence is a crucial prerequisite to improving presence in mixed reality (MR). Traditional presence measures rely on subjective questionnaires, which cannot measure time-varying aspects of presence and are susceptible to user bias. Recent research efforts propose using objective human performance metrics, such as the reaction time between a user's hand-based interaction in response to an explicit cue, as a presence measure. The authors demonstrate that reaction time and presence are correlated in an essential step toward objective presence measurements. Unfortunately, combining an explicit cue and hand interaction to measure reaction time will not suit all scenarios. Additional approaches that leverage implicit cue-interaction mechanisms, such as eye gaze tracking and voice interactions, need to be investigated.

In this work, we outline alternative mechanisms to calculate reaction time using implicit cues, present a fundamental methodology for assessing these mechanisms, and discuss the potential challenges in their evaluation and practical use.

**Index Terms:** Mixed Reality, Presence, Questionnaires

## 1 Introduction

In Mixed Reality (MR), creating a sense of presence is essential to providing an immersive and engaging experience. Presence, the psychological sensation of *being there*, allows users to feel fully absorbed in a virtual environment (VE). Measuring presence is essential for improving it, and traditional approaches have primarily relied on subjective questionnaires. However, presence questionnaires cannot measure the time-varying qualities of presence and can produce unstable, inconsistent, and irreproducible responses due to the prior experience of the participants.

Broader work on assessing presence has explored alternative approaches, including (i) behavioral responses, such as postural, hand-eye coordination, and startle responses; (ii) physiological responses, such as a change in heart rate, skin conductance, and body temperature; and (iii) human performance metrics, such as reaction time and eye-gaze tracking. An ideal *presence* measure is objective, quantitative, not consciously affected by the participant and experimenter, and could be used at runtime without interfering with the virtual experience. It should take advantage of existing interactions in the virtual scene and measure *presence* through these interactions rather than making additional external interventions.

Recent research has demonstrated that the presence and objective human performance measures, such as reaction time [1, 2], are correlated. The authors provide an explicit cue to the user for a response and measure the reaction time between the cue and the hand interaction performed by the user. While this approach satisfies most of the criteria above, it still requires an explicit cue and a user's hand-based interaction. Unfortunately, this intrusive setup may not be practical for all applications, as some may use different interaction mechanisms, such as eye-gaze-based and voice interactions, and have tasks like collaborative sensemaking, which prohibit

priming tasks behavior [9]. Therefore, there is a need to investigate alternative and implicit options that allow estimating reaction time.

In addition to the hand-based interaction exploited by prior work, modern headsets continuously track additional user attributes, such as eye-gaze tracking, hand movement tracking, pose estimation, and voice recognition. These attributes can be used but require extensive investigation into *what will constitute a cue*, *how to establish if the user has responded*, and *how to calculate the reaction time*. These challenges are compounded if the cue is implicit and reaction time is to be measured in routine application interactions. To solve the challenges above, we make the following contribution:

1. Identify inherent interaction mechanisms that allow measuring reaction time implicitly without an explicit cue.
2. Present a methodology to evaluate reaction time estimated using implicit cues and inherent interaction approaches.
3. Outline potential challenges in evaluating the alternative approaches and employing them in practical applications.

## 2 Background and Related Work

The state-of-the-art method of measuring presence uses subjective questionnaires that participants answer after engaging in an MR environment. The most commonly used questionnaires include the Witmer & Singer Presence Questionnaire (PQ) [8], the Igroup Presence Questionnaire (IPQ) [6], and the Slater-Usch-Steed Questionnaire (SUS) [7]. They assess a user's ability to control the environment, the extent to which the environment feels real, the consistency of the virtual world and lived experiences, and other factors that impact presence. However, questionnaires can only measure presence after experience and cannot capture time-varying aspects of presence, denying real-world use to measure in-situ presence.

A large body of work has suggested human performance metrics, such as task completion time, error rate, and actions performed, can be used to indicate presence. Fortunately, modern MR headsets, such as HoloLens 2 and Apple Vision Pro, employ many sensors that capture the user's fine-grained movements and expose them through well-defined interfaces. For example, HoloLens 2 exposes hand tracking, user pose, eye gaze tracking, and voice recognition data, which indicates various aspects of human performance. The ease of access enables the potential use of these metrics as a presence measure in real-time in an unbiased manner.

## 3 Approach

Prior work [1, 2] used criteria defined by Insko et al. [5] to design their user study, which states that a useful measure should be sensitive to different levels of presence, reliably provide repeatable results, correlate with existing presence measures, and should be free from participant's and experimenter's bias. In this section, we revisit the design challenges of a user study shown in Figure 1 for the case studies of *eye tracking* [3] and *voice interactions* [4], both involving implicit cues and hands-free interaction mechanisms.

**1. How do we induce different levels of presence while minimizing confounding variables?** To assess the effect of presence on a metric, we need to induce varying levels of presence controllably. Prior work exploited technical factors [1], such as realism and the plausibility of virtual objects, and human factors [2], such as conditioning. While other factors can also impact presence, the investigations of eye tracking and voice interactions can manipulate the same factors to vary presence.

For eye tracking, we can manipulate presence by varying objects' visual fidelity and behavior within the area of interest. For example, more realistic and contextually appropriate environmental changes, such as a lifelike animation of a painting changing color in an art gallery, may induce a higher level of presence.

For voice interactions, the complexity and relevance of the virtual assistant's prompts can be varied. To alter presence, the virtual assistant can vary the question's complexity from generic to relevant but detailed, such as "What color is the painting?" to "Can you describe the main features of the painting in front of you?".

**2. How do we establish and estimate the baseline presence?** Prior work had to use SUS, PQ, and IPQ questionnaires to measure the ground truth presence, as no other widely accepted presence measures exist. Using questionnaires, the authors had to assess the level of presence at the end of experiments and compare it with aggregate statistics for the reaction times during the experiment.

Alternative approaches, such as eye tracking and voice interactions, can leverage the same approach. The objective but validated questionnaires can establish baseline presence levels for each participant, providing a reference point for subsequent measurements. This method accurately compares traditional subjective assessments and the new implicit cue-based presence measures.

**3. How do we define an implicit cue, record an interaction, and calculate reaction time?** Prior work used explicit cues, such as a change of object color, to prompt an interaction from the user. Implicit cues can also use a similar approach, such as the appearance of a new object, a change in an object's state (e.g., color change), or a verbal prompt from a virtual assistant. However, the cue must be naturally integrated events within the virtual environment. Additionally, hand-based interaction makes it easy to register a user's response. Reaction time is calculated as the difference between the response and cue timestamps. However, answering these questions for eye tracking and voice interactions is not straightforward.

For eye tracking, the system can define areas of interest to monitor the user's gaze coordinates and dwell time [3]. When an implicit cue occurs, such as a new object appearing or a change within the area of interest, the system can detect the first fixation and record the timestamp. Reaction time can be calculated as the difference between the event trigger and the fixation timestamps.

For voice interactions, the virtual assistant can deliver prompts that vary in complexity and relevance. The voice recognition system can employ real-time speech recognition and natural language processing algorithms, using acoustic models to interpret the timing and semantic models to understand the content [4]. Reaction time can be calculated as the difference between the event trigger timestamp (when the prompt is issued) and the timestamp when the system recognizes the user's response as received.

## 4 Discussion

While the proposed methods of using eye tracking and voice interactions to measure presence through implicit cues show promise, several challenges and limitations need to be addressed:

**Variability in user responses.** For eye tracking, different users may exhibit varying gaze patterns and fixation behaviors, affecting the consistency of reaction time measurements. This can be addressed by employing adaptive algorithms that account for individual differences and using personalized calibration procedures. In

voice interactions, speech recognition systems may struggle with accents, speech impediments, or background noise. Mitigating this involves incorporating robust noise-cancellation techniques and training the system using a diverse dataset.

**Calibration and precision.** Eye tracking systems must be accurately calibrated for each user and maintained through regular recalibration and high-precision sensors. Voice recognition systems need precise tuning and continuous learning algorithms to adapt to the user's voice over time, improving recognition accuracy.

**Context sensitivity.** For eye tracking, the context in which the area of interest is defined can influence gaze behavior. This can be managed with adaptive definitions for the area of interest that adjust based on user behavior and the environment's context. For voice interactions, prompts must be engaging yet not overly complex, achieved through adaptive dialogue systems that modify prompt complexity based on real-time user feedback.

**Real-time processing and seamless integration.** Eye tracking and voice interaction must process data in real-time, requiring low latency and high processing speed. Utilizing edge computing and optimization algorithms can address this need. Additionally, robust software and hardware integration, standardized APIs, and compatibility with different MR platforms are necessary to ensure smooth operation without disrupting the user experience.

Note that eye tracking and voice interactions enable a more inclusive and practical use of reaction time as a presence measure. Also, using a cue and interaction combination does not need to be static, and the application developer can choose among these mechanisms based on the context, user attributes, and application.

## 5 Conclusion

Traditional subjective questionnaires fall short of capturing the dynamic nature of presence, and hand-based interaction made a significant stride toward objective presence measurement. Leveraging natural interactions offers a non-intrusive and objective measure. This paper discussed the potential of using eye tracking and voice interactions with implicit cues to measure presence in MR. Despite their promise as reliable presence measures, they require further research to address existing challenges and ensure practical application, leading to effective and user-friendly MR experiences.

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